Stratazeal 2.0

ROUND - 2 SUBMISSION

The Case Quest : Symbiosis Centre for Management and Human Resource Development (SCMHRD)

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Track: Business Analytics

Fraud Detection: Key Risk Factors and Patterns

Analyzing transaction data to uncover key drivers, high-risk time windows, regional/credit patterns, and customer risk profiles influencing fraud.

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Situation

Current scenario of the company

Rapid growth in digital transactions

Increasing fraud cases during certain hours & channels.

Manual fraud monitoring, not scalable.

QUESTIONS?

Which **factors (numerical & categorical)** have the strongest link with fraud?

How **temporal & channel behavior** (time of transaction × channel) influences fraud patterns?

How **specific risk variables** (Region Risk Score, Credit Score, Avg Transaction Amount) relate to fraud?

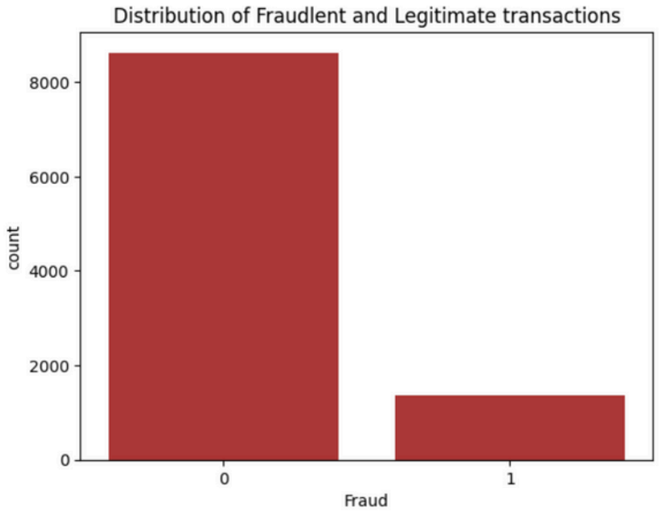
Who is more likely to commit fraud given **real customer scenarios**?

Dataset Overview

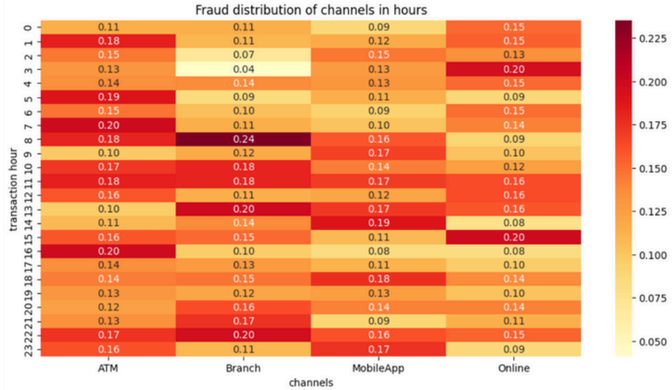
- 10,000 transactions analyzed.
- 19 key features (Age, Location, Account Type, Transaction Amount, Channel, etc.).
- **Fraud Rate: 13.77%** (imbalanced data → oversampling/under sampling suggested for modeling)
- Negative transactions flagged and converted to absolute.
- **Categorical variables encoded** (One-Hot Encoding).
- Numerical & categorical features analyzed separately

Key Findings from EDA

Feature Type	Key Observation	Link to Fraud
Numerical	Region Risk Score, Credit Score, Avg Transaction Amount	Weak correlation (r ≈ 0, p > 0.4)
Categorical	Channel, Account Type, Txn Frequency Band, Location	Strong fraud indicators: ATM, Savings, Urban/Semi-Urban, Low-frequency txns
Temporal Patterns	Fraud spikes at certain hours	ATM: 5–8 AM, 4 PM, 10–11 PM; Branch: 8 AM



Confirms that **fraud is relatively rare**, so automated monitoring should focus on high-risk patterns rather than overall volume.



Enables **focused real-time monitoring** for specific channels during high-risk hours → **efficient fraud prevention.**

Fraud Risk Scoring & Case Resolution

Build a structured fraud risk scoring system to identify and rank high-risk customers, enabling proactive fraud prevention and smarter resource allocation.

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Fraud Risk Score Framework

Past fraud, txn behaviour, odd hours, risky channels, credit & literacy.

Scoring Scale:

- Low Risk = 0–4
- Medium Risk = 5–8
- High Risk = 9+

Risk Comparison: Fraud Risk Score by Person

Person	Risk Level	Why Risky? (Key Triggers)
Person 1	High (11)	Past Fraud, Odd hour (2:30 AM), Online Channel, High transaction vs avg, Low literacy
Person 2	Medium (6)	ATM usage, Higher transaction vs avg, Past Fraud
Person 3	Low (3)	Branch (in-person), Normal transaction size, Higher literacy

Executive Insights

- **Person 1** is the most likely fraudster (Score = 11, High Risk)
- **Key Fraud Drivers**: Night-time online transactions + low literacy + past history.
- **High false positives** damaging customer experience.

Company-Specific Pain Points

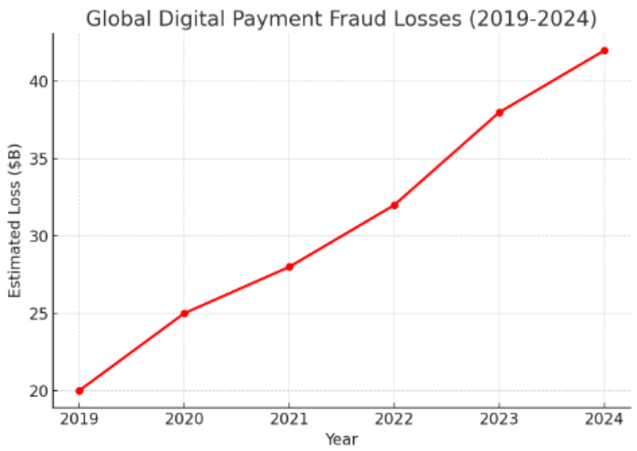
- ATM transactions show disproportionately **higher fraud rates** in early mornings (5–8 AM) and late evenings.
- Features like **Credit Score, Region Risk Score, and Avg. Transaction Amount** show almost no correlation with fraud in this dataset.
- **High false positives** → good customers flagged, leading to churn.
- Fraud "**windows**" (e.g., ATM at dawn, Semi-urban branches at 1 PM) are not tracked in real-time dashboards.

Initiative Needed

- **Leverage advanced analytics** (EDA, ML models) to uncover hidden fraud patterns.
- **Identify high-risk time windows** & channels for targeted monitoring.
- Build **customer risk profiling** to reduce false positives.
- **Automate fraud alerts** → freeing manual teams for only edge cases.

Industry Challenge

- **Global Scale**: Fraud in digital payments estimated to cause **$40B+ losses annually (2023-24 reports)**.
- **Evolving Tactics**: Fraudsters use AI-driven social engineering & synthetic identities.
- **Customer Impact**: Loss of trust → higher churn & reputational damage.
- **Regulatory Pressure**: Industries face stricter compliance (KYC, AML, RBI/PCI-DSS standards).
- **Cost of Detection:** Balancing fraud prevention with seamless customer experience is difficult.



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Thank you